



$$\frac{2}{3} \div \frac{2}{3} = 1$$

$$\sqrt[n]{\frac{y_n - 5}{2}}$$

$$a > \sqrt[n]{\frac{n-5}{5}}$$

$$a_n \rightarrow a$$



$$(\sqrt{a_n})^2 = (\sqrt{a})^2 = a$$

一、二、三、四、五、六、七、八、九、十、十一、十二、十三、十四、十五、十六、十七、十八、十九、二十、二十一、二十二、二十三、二十四、二十五、二十六、二十七、二十八、二十九、三十、三十一、三十二、三十三、三十四、三十五、三十六、三十七、三十八、三十九、四十、四十一、四十二、四十三、四十四、四十五、四十六、四十七、四十八、四十九、五十、五十一、五十二、五十三、五十四、五十五、五十六、五十七、五十八、五十九、六十、六十一、六十二、六十三、六十四、六十五、六十六、六十七、六十八、六十九、七十、七十一、七十二、七十三、七十四、七十五、七十六、七十七、七十八、七十九、八十、八十一、八十二、八十三、八十四、八十五、八十六、八十七、八十八、八十九、九十、九十一、九十二、九十三、九十四、九十五、九十六、九十七、九十八、九十九、一百。

Can

$$x_{a_1} + x_{b_1} + x_{c_1} + x_{d_1} + x_{e_1} + x_{f_1} + x_{g_1} + x_{h_1} + x_{i_1} + x_{j_1} + x_{k_1} + x_{l_1} + x_{m_1} + x_{n_1} + x_{o_1} + x_{p_1} + x_{q_1} + x_{r_1} + x_{s_1} + x_{t_1} + x_{u_1} + x_{v_1} + x_{w_1} + x_{x_1} + x_{y_1} + x_{z_1}$$

20

$$\sqrt{5} = 2$$

مجلس

$$C \leq \frac{1}{\sqrt{n}} \left| \frac{a_n - a}{\sqrt{a_n - a}} \right|$$

10.7h.G

$$d_{\text{apmex}} Th - G_{\text{apmex}} \sqrt{a_{\text{apmex}}} - \sqrt{a} = 0$$

10-11-19

Ex 3.1

$$v) \frac{1}{2} \ln \left(\frac{1}{2} \right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0 \quad \text{et } \frac{1}{n} \rightarrow 0$$

196-1172

$$3 - \frac{4 - 3}{2} = 3$$

طوبى

$$z = \left(\frac{1}{2} - \frac{1}{2}i \right)$$

$$\left(-\frac{3}{2}\right)^n - 56 - \text{ch naig en}$$

$$\lim_{n \rightarrow \infty} u_n = 0$$

$$\frac{1}{5} \log \frac{L(\frac{1}{5})}{L(\frac{1}{5}+1)}$$

$$5) U_n = \frac{3^n + 5}{5 \cdot 7 \cdot 11 + 1} \cdot \frac{3^n - 5^n - 1}{\left(\frac{1}{2}\right)^n} \cdot \frac{3^n - 5^n - 1}{\left(\frac{1}{2}\right)^n}$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{x^2} - \frac{1}{x}}{\frac{1}{x} - 1} = 0$$

5

$$4) U_n = S^n - 1 = S^n \left[1 - \left(\frac{1}{S} \right)^n \right]$$

$$b) u_n = \frac{2n^2 - 3n + 1}{n^2 - 2n - 15} = \frac{(n-1)(2n+1)}{(n-5)(n+3)}$$

$$7) \quad u_n = \sqrt{n+1} \cdot n$$

$$\lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 - n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + 11} + n}$$

$$\lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 + n}) - n}{\sqrt{n^2 + n} + n}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 + n - 1}{n \sqrt{n^2 + 1}}$$

$$\frac{1}{2}$$

Full page



$$= \lim_{n \rightarrow +\infty} \frac{(\sqrt{n^2 + n})^2 - n^2}{\sqrt{n^2 + n} + n} =$$

$$= \lim_{n \rightarrow +\infty} \frac{n^2 + n - n^2}{\sqrt{n^2 + n} + n} = \lim_{n \rightarrow +\infty} \frac{n}{\sqrt{n^2 + n} + n}$$

$$= \lim_{n \rightarrow +\infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{2}$$

$$= \lim_{n \rightarrow +\infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{2}$$

$$= \frac{1}{2}$$

$$② \lim_{n \rightarrow +\infty} \frac{n - (-1)^n}{n^3 + 1} =$$

$$= \frac{1 - (-1)^n}{15 - 1} = \frac{1 - (-1)^n}{14}$$

$$u_{n+1} > u_n = \frac{3u_n + 4}{u_n + 1}$$

$$u_{n+2} > u_{n+1} \Rightarrow n \text{ c.v.}$$

(3) (u_n)

$$l = \sqrt{3l + 4} \Rightarrow l^2 = 3l + 4$$

$$\Delta = (-3)^2 + 16 = 25$$

$$l_1 = \frac{3 + 5}{2} = 4 \checkmark$$

$$l_2 = \frac{3 - 5}{2} = -1$$

$$110 > 2 \cdot 34$$

$$a_n = \frac{(a_{n-1})^2 + 1}{2}$$

$$l = l_{+} a_n \Rightarrow l = l_{+} a_{n-1}$$

$$l = \frac{l^2 + 1}{2} \Rightarrow 2l - l^2 - 1 = 0$$

$$\Delta = (2)^2 - 4 = 0$$

$$l_1 = l_2 = \frac{-2}{-2} = 1$$

$$\lim_{n \rightarrow +\infty} a_n = 1$$

2^{es} methode:

Montrons que a_n
par 1.

$a_n < 1$
pour $n=0$
Supposons que
Montrons q



$$n \left(1 + \frac{\ln n}{n} \right) \leq 1$$

$$\frac{1}{\left(1 + \frac{\ln n}{n} \right)} \leq 1$$

$$\frac{1}{1 + \frac{\ln n}{n}} \leq 1$$

$$\Rightarrow \lim_{n \rightarrow +\infty} \frac{1}{1 + \frac{\ln n}{n}} = 1$$



④

$$\sum \frac{P}{k(k+1)(k+2)}$$

$$\frac{P}{k(k+1)(k+2)} = \frac{a}{k} + \frac{b}{k+1} + \frac{c}{k+2}$$

① $\sum_{k=0}^{+\infty} \frac{2}{5^k} = 2 \sum_{k=0}^{+\infty} \left(\frac{1}{5}\right)^k$

$\left(\frac{1}{5}\right)^k$ S.G. de la suite $q = \frac{1}{5} < 1$

② $\sum_{k=0}^{+\infty} \frac{5^k}{3^k} = \sum_{k=0}^{+\infty} \left(\frac{5}{3}\right)^k$

$\left(\frac{5}{3}\right)^k$ S.G. de la suite $q = \frac{5}{3} > 1$

③ $\sum_{k=0}^{+\infty} \frac{2k+3}{4+6k}$; $u_k = \frac{2k+3}{4+6k}$

$\lim_{k \rightarrow +\infty} u_k = \lim_{k \rightarrow +\infty} \frac{2k}{6k} = \frac{2}{6} = \frac{1}{3} \neq 0 \Rightarrow \sum u_n$ D.V.

$\Rightarrow l = \frac{l^2}{2}$
 $\Delta = (2)^2$
 $l_1 = l_2 = \dots$

$\lim_{n \rightarrow +\infty} a_n =$

2nd meth

Montrons que
 par 1.

$a_n <$
 pour $n=0$

Supposons que
 Montrons que

supposons que $\lim_{n \rightarrow +\infty} u_n = l$

une suite Convergente vers l alors
 la suite v_n de $\lim$ Convergent vers
 la même l

$$u_n = \cos \left(\frac{2\pi (17n+1)}{17} \right)$$

$$n \in \mathbb{N}$$

$$= \cos \left(\frac{2\pi (17n)}{17} + \frac{2\pi}{17} \right)$$

$$u_{17n} = \cos \left(\frac{2\pi (17n)}{17} \right) = \cos(2\pi n) = 1$$

$$k \text{ pair } \cos(k\pi) = 1$$

$$k \text{ impair } \cos(k\pi) = -1$$

$$u_{17n+1} = \cos \left(\frac{2\pi (17n)}{17} + \frac{2\pi}{17} \right)$$



$$\lim_{n \rightarrow \infty} \ln x \ln n \rightarrow 0$$

$\sum_{k=2}^n \ln k$

$$\ln n = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots$$

$$= \sum_{k=2}^n \ln k$$

$$1 \leq k \leq n$$

$$0 \leq \ln k \leq \ln n$$

$$n \leq n \ln k \leq n \ln n$$



$$u_{n+1} = \sqrt{3u_n + 4}$$

$$(1) u_n < 4$$

pour $n=1$: $u_1 = 2 < 4$

Supposons que: $u_n < 4$

Montrons que: $u_{n+1} < 4$

$$u_n < 4 \Rightarrow 3u_n < 12 \Rightarrow 3u_n + 4 < 16$$

$$\Rightarrow \sqrt{3u_n + 4} < \sqrt{16} = 4$$

$$\Rightarrow u_{n+1} < 4$$

$$(2) \text{ On a: } u_2 = \sqrt{3u_1 + 4} = \sqrt{3 \times 2 + 4} = \sqrt{10} > 2$$

Supposons que $u_{n+1} > u_n$

Montrons que: $u_{n+2} > u_{n+1}$

$$u_{n+1} > u_n \Rightarrow 3u_{n+1} + 4 > 3u_n + 4$$

$$\Rightarrow u_{n+2} > u_{n+1}$$

et majorée $\Rightarrow u_n$ c.v.

$$l = \sqrt{3l + 4} \Rightarrow l^2 - 3l - 4 = 0$$

$$\Delta = (-3)^2 + 16 = 25$$

$$l_1 = \frac{3+5}{2} = 4$$

$$l_2 = \frac{3-5}{2} = -1$$

Exot: $\lim_{n \rightarrow \infty} u_n$ avec $u_n = \frac{(a_n)^2 + 1}{2}$

$$a_n \rightarrow 0$$



$$\Rightarrow \frac{7-3n}{6n^2+2} < 0 < \frac{7+3n}{6n^2+2}$$

$\swarrow$ $\searrow$ $\swarrow$ $\searrow$
 0 0

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

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$$U_n = \frac{1}{n^2}$$

$$U_n = \frac{a \cdot b}{e^{u(n)}} = \frac{a \cdot b}{e^{\ln(a^b)}} = \frac{a \cdot b}{e^{b \ln a}} = \frac{a \cdot b}{e^{b \ln a - 2 \ln a}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$



$$L_n = \frac{3^n + 5^n}{7^n - 6^n} = \frac{5^n}{7^n} \left[\left(\frac{3}{5} \right)^n + 1 \right] \left[1 - \left(\frac{6}{7} \right)^n \right]$$

$$= \left(\frac{5}{7} \right)^n \times \left[\left(\frac{3}{5} \right)^n + 1 \right] \left[1 - \left(\frac{6}{7} \right)^n \right] \rightarrow 0$$

6

$$L_n = \frac{9n^2 - 5n + 2}{n^2 - 4n}$$

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \frac{9n^2 - 5n + 2}{n^2 - 4n} = \lim_{n \rightarrow \infty} \frac{9 - \frac{5}{n} + \frac{2}{n^2}}{1 - \frac{4}{n}} = \frac{9}{1} = 9$$

$$\lim_{n \rightarrow \infty} \frac{9n^2 - 5n + 2}{n^2 - 4n} = 9$$

7) $L_n = \sqrt{n^2 + n} - n$

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 + n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + n} + n}$$



$$u_{n+2} - 3u_{n+1} + 4 > 3u_n + 4$$

$$\Rightarrow \sqrt{3u_{n+1} + 4} > \sqrt{3u_n + 4}$$

$$u_{n+2} > u_{n+1}$$

et majoré $\Rightarrow u_n$ est majoré

$$l = \sqrt{3l + 4} \Rightarrow l^2 - 3l - 4 = 0$$

$$\Delta = (-3)^2 + 16 = 25$$

$$l_1 = \frac{3+5}{2} = 4 \checkmark$$

$$l_2 = \frac{3-5}{2} = -1$$

$$2+4 = \sqrt{10} > 2 \text{ لا}$$

Est: $\lim_{n \rightarrow \infty} u_n = a_n$

$$a_1 = 0$$

$$l = \lim_{n \rightarrow \infty} a_n \Rightarrow l = \lim_{n \rightarrow \infty} \frac{a_n^2 + 1}{2}$$

$$\Rightarrow l = \frac{l^2 + 1}{2} \Rightarrow 2l - l^2 - 1 = 0$$

$$\Delta = (-2)^2 - 4 = 0$$

$$l_1 = l_2 = \frac{2}{-2} = -1$$

$$\lim_{n \rightarrow \infty} u_n = 1$$

2^e méthode:

Montrons que a_n majoré

$$a_0 = 1$$

$$a_n < 1$$

pour $n=0$ $a_0 = 0 < 1$ vrai

Supposons que $a_n < 1$

Montrons que $a_{n+1} < 1$

$$a_n < 1$$

$$\Rightarrow a_n^2 + 1 < 2 \Rightarrow \frac{a_n^2 + 1}{2} < 1$$

$$\Rightarrow a_{n+1} < 1$$

$$a_{n+1} - a_n = \frac{a_n^2 + 1}{2} - a_n$$

$$= \frac{a_n^2 + 1 - 2a_n}{2} = \frac{(a_n - 1)^2}{2} > 0$$



$$\sqrt{(-1)^n n^2} \leq n^2$$

$$n - n^2 \leq n + (-1)^n n^2 \leq n + n^2$$

$$\Rightarrow \frac{n - n^2}{n^3 + 1} \leq u_n \leq \frac{n + n^2}{n^3 + 1}$$

$$\Rightarrow \lim_{n \rightarrow +\infty} u_n = 0$$

$$(9) \quad u_n = \frac{7 + 3n \cos(n^2 + r)}{6n^2 + 2}$$

$$\Rightarrow -1 \leq \cos(n^2 + r) \leq 1$$

$$\Rightarrow -3n \leq 3n \cos(n^2 + r) \leq 3n$$

$$\Rightarrow 7 - 3n \leq 7 + 3n \cos(n^2 + r) \leq 7 + 3n$$



$$\left| \frac{4n-5}{n} \right| < \varepsilon$$

$$\left| \frac{4n-5}{n^2} \right| < \frac{4n}{n^2} = \frac{4}{n}$$

$$\forall \varepsilon > 0 \quad \frac{4}{n} < \varepsilon$$

$$\left| \frac{4n-5}{n^2} \right| < \frac{4}{n} < \varepsilon$$

$$n > \frac{4}{\varepsilon} = n_\varepsilon$$

$$\forall n > n_\varepsilon, \quad \frac{4}{\varepsilon} = n_\varepsilon \Rightarrow \frac{4}{\varepsilon} = n_\varepsilon \Rightarrow 4 \leq n_\varepsilon$$

$$\left| \frac{4n-5}{n^2} \right| < \varepsilon$$

Exo 2:

$$a_n \rightarrow a$$

$$a_n \rightarrow \sqrt{a}$$

d'après Th. G.
 $\lim_{n \rightarrow \infty} (a_n - \sqrt{a}) = 0$

Ex 3:

$$| \sqrt{a_n} - \sqrt{a} | = | \sqrt{a_n + \sqrt{a}} - \sqrt{a} |$$

$$| (\sqrt{a_n})^2 - (\sqrt{a})^2 | = | a_n - a |$$

$$| \sqrt{a_n} - \sqrt{a} | = \frac{| a_n - a |}{| \sqrt{a_n} + \sqrt{a} |} < \frac{| a_n - a |}{\sqrt{a}}$$

Com

$$\sqrt{a_n} + \sqrt{a} > \sqrt{a}$$

$$\frac{1}{\sqrt{a_n} + \sqrt{a}} < \frac{1}{\sqrt{a}}$$

$$\frac{| a_n - a |}{\sqrt{a_n} + \sqrt{a}} < \frac{| a_n - a |}{\sqrt{a}}$$

$$c \leq | a_n - a | < \frac{c}{\sqrt{a}}$$

$$\frac{c}{\sqrt{a}} > 0 \quad \text{Th. G.}$$

2

$$u_n = 1 + \frac{1}{2^n} \quad \text{S.G. } q = \frac{1}{2} < 1$$

$$2) \quad u_n = \frac{1}{n} + \left(\frac{1}{3} \right)^n$$

$$\left(\frac{1}{3} \right)^n \text{ S.G. de raison } \frac{1}{3}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{1}{3} \right)^n = 0 \quad \text{et } \frac{1}{n} \rightarrow 0$$

$$\Rightarrow u_n \rightarrow 0$$

$$3) \quad u_n = \frac{1 - 3^{1/n}}{2^{1/n} - 1}$$

$$= \frac{e^{1/n} - 1}{2^{1/n} - 1}$$

$$= \frac{1}{1} \cdot \left(\frac{1}{2} \right)^n = \frac{1}{2}$$

$$\left(\frac{1}{2} \right)^n \text{ S.G. de raison } \frac{1}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{1}{2} \right)^n = 0$$

pas de limite

limite n'est pas nulle

$$5) \quad u_n = \frac{3^n + 5^n}{7^n + 9^n} \left[\left(\frac{3}{7} \right)^n + \left(\frac{5}{9} \right)^n \right]$$

$$\left(\frac{3}{7} \right)^n \cdot \left(\frac{5}{9} \right)^n \rightarrow 0$$

Ex

$$4) \quad u_n = 5^n - 4^n = 5^n \left[1 - \left(\frac{4}{5} \right)^n \right]$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{2n^2 - 3n + 2}{n^2} = \lim_{n \rightarrow \infty} \frac{4n - 3}{2n} = 2$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{2n^2 - 3n + 2}{n^2} = \lim_{n \rightarrow \infty} \frac{4n - 3}{2n} = 2$$

$$= 2$$

$$7) \quad u_n = \sqrt{n+1} - n$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{(\sqrt{n+1} - n)(\sqrt{n+1} + n)}{(\sqrt{n+1} + n)} = \lim_{n \rightarrow \infty} \frac{1 - n^2}{\sqrt{n+1} + n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{(\sqrt{n+1} - n) \cdot n^2}{n^2} = \lim_{n \rightarrow \infty} \frac{1 - n^2}{n^2} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1 - n^2}{n^2} = 0$$

$$= \frac{1}{2}$$



$$K\pi \rightarrow \alpha + \pi$$

$$= \pi \left(\frac{\pi}{6} + \frac{1}{n} \right)$$

$f(n)$ continue : $\lim_{n \rightarrow \infty} f(n) =$

$$\Rightarrow \lim_{n \rightarrow \infty} f(n) = f\left(\lim_{n \rightarrow \infty} n\right) = f(\infty)$$

Comme $\sin$ est une fonction continue

a-lors $\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{6} + \frac{1}{n}\right) = \sin\left(\frac{\pi}{6}\right)$

$$\sin\left(\lim_{n \rightarrow \infty} \left(\frac{\pi}{6} + \frac{1}{n}\right)\right) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

Exercice 21

$$\lim_{n \rightarrow \infty} \cos\left(\frac{2n\pi}{n}\right) \text{ et } \lim_{n \rightarrow \infty} \frac{1}{n}$$

Preuve $n \geq 1$



$$= \cos\left(\frac{2\pi n + 17}{17}\right) + \dots$$

$$= \cos\left(2\pi n + \frac{2\pi}{17}\right)$$

$$= \cos\left(\frac{2\pi}{17}\right) \neq 1$$

$$2\pi n + \alpha = \alpha$$

$(u_n + v_n)$ n'admet pas de limite

$$-1 \leq \cos\left(\frac{2\pi n}{17}\right) \leq 1$$

$$\Rightarrow \frac{-1}{n} \leq \frac{1}{n} \cos\left(\frac{2\pi n}{17}\right) \leq \frac{1}{n}$$

$\frac{-1}{n} \xrightarrow[n \rightarrow \infty]{} 0$ $\frac{1}{n} \xrightarrow[n \rightarrow \infty]{} 0$



① $\sum u_n$ avec u_n S.G.
de raison q :

si $q \leq -1$ n'admet pas $\sum u_n$
 $-1 < q < 1 \Rightarrow \lim_{n \rightarrow +\infty} u_n = 0 \Rightarrow \sum u_n$ D.V.
 $q > 1 \Rightarrow \lim_{n \rightarrow +\infty} u_n = +\infty \Rightarrow \sum u_n$ D.V.

$$u_{n+1} > u_n = 3u_n$$

$$u_{n+2} =$$

② Condition nécessaire, q

$$\sum u_n \text{ C.V.} \Rightarrow \lim_{n \rightarrow +\infty} u_n = 0$$

$$\Rightarrow \lim_{n \rightarrow +\infty} u_n \neq 0 \Rightarrow \sum u_n \text{ D.V.}$$

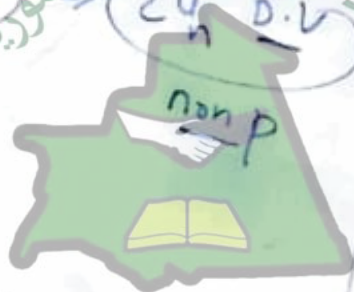
③ Alambert

$$l = \lim_{n \rightarrow +\infty} \frac{u_{n+1}}{u_n}$$

$$l > 1 \Rightarrow \sum u_n \text{ D.V.}$$

$$l < 1 \Rightarrow \sum u_n \text{ C.V.}$$

$$l = 1$$



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③ (u_n) et major

$$l = \sqrt{3l+4} \Rightarrow l^2 =$$

$$\Delta = (-3)^2 + 16 = 25$$

$$l_1 = \frac{3+5}{2} = 4 \checkmark$$

$$l_2 = \frac{3-5}{2} = -1$$

EX 7: $\lim_{n \rightarrow +\infty} n^n$ avec: $q_n =$
 $a_1 = 0$



$$\lim_{n \rightarrow +\infty} \frac{3\sqrt{n}}{5\sqrt{n}+5} = \frac{3}{5}$$

Soit $\varepsilon > 0$

$$\left| \frac{3\sqrt{n}}{5\sqrt{n}+5} - \frac{3}{5} \right| < \varepsilon$$

à d
 $\exists n_\varepsilon \in \mathbb{N} / \forall n \geq n_\varepsilon$

$$|u_n - \frac{3}{5}| < \varepsilon$$

$$|u_n - \frac{3}{5}| < \varepsilon$$

$\varepsilon > 0$ tq

$$\varepsilon \Rightarrow \left| \frac{n}{n+1} - 1 \right| < \varepsilon$$

$$< \varepsilon =$$

$$\left| \frac{12\sqrt{n} - 12\sqrt{n} - 15}{16\sqrt{n} + 20} \right| < \varepsilon$$

$$\left| \frac{-15}{16\sqrt{n} + 20} \right| < \varepsilon$$

$$\Rightarrow \frac{15}{16\sqrt{n} + 20} < \varepsilon$$

$$\Rightarrow 16\sqrt{n} + 20 > \frac{15}{\varepsilon}$$

$$| < \varepsilon \Rightarrow \frac{1}{n+1} < \varepsilon$$

$$\frac{1}{a} < \frac{1}{b}$$

$$> \frac{1}{\varepsilon} \Rightarrow n > \frac{1}{\varepsilon} - 1 \Rightarrow n > \left(\frac{1}{16} \left(\frac{15}{\varepsilon} - 20 \right) \right)^2$$

$$n_\varepsilon = \frac{1}{\varepsilon} - 1$$

$$\forall n \geq n_\varepsilon \Rightarrow \frac{1}{n} < \frac{1}{n_\varepsilon} < \varepsilon$$

$$\text{on pose } n_\varepsilon = \left\lceil \frac{1}{16} \left(\frac{15}{\varepsilon} - 20 \right) \right\rceil$$

$$\forall \varepsilon > 0 \exists n_\varepsilon = \left\lceil \frac{1}{16} \left(\frac{15}{\varepsilon} - 20 \right) \right\rceil$$

$$\forall n \geq n_\varepsilon$$

$$\left| u_n - \frac{3}{5} \right| < \varepsilon$$



$$\frac{1}{n+1} \leq \frac{1}{n+2k} \leq \frac{1}{n}$$

$$\frac{1}{n} \leq \frac{1}{n+2k} \leq \frac{1}{n}$$

$$\frac{1}{n} \leq \frac{1}{n+2k} \leq \frac{1}{n}$$

$$\frac{1}{n} \leq \frac{1}{n+2k} \leq \frac{1}{n} \quad \sum_{k=1}^n \frac{1}{n+2k} \leq \frac{1}{n}$$

$$\sum_{k=p}^n \frac{1}{n+2k} \leq \frac{1}{n-p+1}$$

$$\frac{1}{n+2k} \leq \frac{1}{n-p+1} \leq \frac{1}{n}$$

Exemple

$$\begin{cases} u_1 = 2 \\ u_n = \sqrt{3u_{n-1} + 4} \end{cases}$$

① $u_n \leq 4$

pour $n=1: u_1 = 2 < 4$ vraie

Supposons que: $u_n < 4$

Montrons que: $u_{n+1} < 4$

$$u_n < 4 \Rightarrow 3u_n < 12 \Rightarrow 3u_n + 4 < 16$$

$$\Rightarrow \sqrt{3u_n + 4} < \sqrt{16} = 4$$

$$\Rightarrow u_{n+1} < 4$$

2. On a: $u_2 = \sqrt{3u_1 + 4} = \sqrt{3 \cdot 2 + 4} = \sqrt{10} > 2 = u_1$

Supposons que $u_n > u_{n-1}$

Montrons que $u_{n+1} > u_n$

$$u_n > u_{n-1} \Rightarrow 3u_n + 4 > 3u_{n-1} + 4$$

$$\Rightarrow \sqrt{3u_n + 4} > \sqrt{3u_{n-1} + 4}$$

$$u_{n+1} > u_n$$

③ (u_n) est majorée $\Rightarrow u_n$ CV

$$l = \sqrt{3l + 4} \Rightarrow l^2 - 3l - 4 = 0$$

$$\Delta = (-3)^2 - 4 = 5$$

$$l_1 = \frac{3 + \sqrt{5}}{2} \approx 2.618$$

$$l_2 = \frac{3 - \sqrt{5}}{2} \approx -0.618$$

le point fixe est $l = \frac{3 + \sqrt{5}}{2}$

$$l = \lim_{n \rightarrow \infty} u_n \Rightarrow l = \lim_{n \rightarrow \infty} \sqrt{3u_{n-1} + 4}$$

$$\Rightarrow l = \sqrt{3l + 4} \Rightarrow l^2 - 3l - 4 = 0$$

$$\Delta = (-3)^2 - 4 = 5$$

$$l_1 = \frac{3 + \sqrt{5}}{2} \approx 2.618$$

$$l_2 = \frac{3 - \sqrt{5}}{2} \approx -0.618$$

2^e méthode:

Montrons que u_n majorée par 1.

$$u_n < 1$$

pour $n=0: u_0 = 0 < 1$ vraie

Supposons que $u_n < 1$

Montrons que $u_{n+1} < 1$

$$u_n < 1 \Rightarrow u_n^2 + 1 < 2 \Rightarrow \frac{u_n^2 + 1}{2} < 1$$

$$\Rightarrow u_{n+1} < 1$$

$$u_{n+1} - u_n = \frac{u_n^2 + 1}{2} - u_n = \frac{u_n^2 - 2u_n + 1}{2} = \frac{(u_n - 1)^2}{2}$$

arrangement d'une

$$\sum_{k=1}^n u_k$$

$$\sum_{k=1}^n u_k$$

$$= + \infty$$

(17) - $U_n = (-1)^n \frac{n^3}{n!}$

$$-1 \leq (-1)^n \leq 1$$

$$\Rightarrow \frac{-n^3}{n!} \leq (-1)^n \frac{n^3}{n!} \leq \frac{n^3}{n!}$$

$$\lim_{n \rightarrow +\infty} \frac{-n^3}{n(n-1)(n-2)(n-3) \dots 1} \leq \lim_{n \rightarrow +\infty} \frac{n^3}{n(n-1)(n-2)(n-3) \dots 1}$$

$$\lim_{n \rightarrow +\infty} \frac{1}{(n-3)(n-2)(n-1)} \leq \lim_{n \rightarrow +\infty} \frac{1}{(n-1)} \leq \lim_{n \rightarrow +\infty} \frac{1}{(n-1)}$$